Akeli Adilijiang

CS/CE Student | Orange County, CA
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TECHNICAL SKILLS

Programming: C++ (C++17/23), C, Python, GLSL
Graphics & UI: SDL2/SDL3, OpenGL, Qt, Win32/GDI, Dear ImGui
Tools: Git, GitHub, CMake/CTest, Visual Studio/MSVC, VS Code
Languages: English (Fluent), Chinese (Native), Japanese (Conversational)

PROJECTS

ElysiaEngine | C++23, SDL3, CMake

- Independently developed a modular 2D engine with separate engine and game modules, architecture documentation, and integration examples.
- Designed scene lifecycle and recovery, action-based input routing, and rendering/UI systems for reuse across games.
- Developed CMake/CTest regression suites for input lifecycle safety, UI callbacks, and persistence failure boundaries.
- Maintain the engine and design game-flow systems for Codex Zero, a game in development by the seven-member Dangling Pointer team; also support club Game Jam projects with the engine.

SoftRenderer | C++17, Win32/GDI, CMake

- Built a software 3D rendering pipeline from scratch with an OpenGL-like interface for vertex processing, rasterization, and fragment shading.
- Implemented perspective-correct interpolation, texture sampling, depth testing, blending, and frustum clipping.
- Implemented a custom Dear ImGui backend with indexed drawing, clip-rectangle handling, and render-state restoration.

PathFindingVisualizer | C++17, SDL2, Dear ImGui

- Built an A*, Dijkstra, BFS, and Greedy search visualizer with wall/weight editing, algorithm switching, and search-state snapshots for step playback and rollback.
- Designed an extensible algorithm interface, published a web demo, and used the visualizer for a data structures course demonstration.

8-bit Computer | 74-series TTL, Digital Logic

- Built an 8-bit computer on breadboards with an ALU, program counter, shared bus, and control unit; integrated 16 x 8-bit memory and two 8-bit registers.
- Implemented 8-bit addition, subtraction, and jump instructions, and brought the assembled computer into operation.

TECHNICAL LEADERSHIP

Game Programming and Simulation Club | Saddleback College 2026 - Present
Founder / President

- Founded and lead a club of 30+ members, teaching C++ game development from scratch and organizing small Game Jams for hands-on practice.
- Developed Project Autonomy, a C++23/SDL3 game AI programming competition platform.
- Designed C++ lessons and trained club leaders to teach game development.

ADDITIONAL EXPERIENCE

Saddleback College | CARE Corner Assistant Oct. 2025 - Aug. 2026

- Assisted 70+ students daily with campus resources, service referrals, check-ins, and front-desk support.

CES 2026 | Exhibitor / Booth Staff Jan. 2026

- Delivered live product demonstrations for Ecoldbrew, explained technical features, and answered attendee questions in a high-traffic exhibition environment.

EDUCATION

Saddleback College | Computer Science / Computer Engineering GPA: 3.87
Expected Transfer: Aug. 2027
Honors: NSF S-STEM Scholarship Recipient - 2025, 2026
Relevant Coursework: Data Structures, Physics (E&M), Linear Algebra, Calculus